

Chapter 31

Fungi

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: Mighty Mushrooms

- **Fungi** are diverse and widespread.
- They are essential for the well-being of most terrestrial ecosystems because they **break down organic material** and **recycle vital nutrients**.
- Fungi are **heterotrophs** and **absorb** nutrients from outside of their body.
- Fungi use enzymes to break down a large variety of complex molecules into smaller organic compounds.
- The versatility of these enzymes contributes to fungi's ecological success.

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- The most common body structures are **multicellular filaments** and **single cells (yeasts)**.
 - Some species grow as either filaments or yeasts; others grow as both.
 - Fungi exhibit diverse lifestyles:
 - *Decomposers / saprophytes*
 - *Parasites + -*
 - *Mutualists + +*

Fungal Morphology : hyphae

- The morphology of multicellular fungi enhances their ability to absorb nutrients.
- Fungi consist of **mycelia**, **networks** of branched **hyphae** adapted for absorption.
- Most fungi have **cell walls** made of **chitin**.

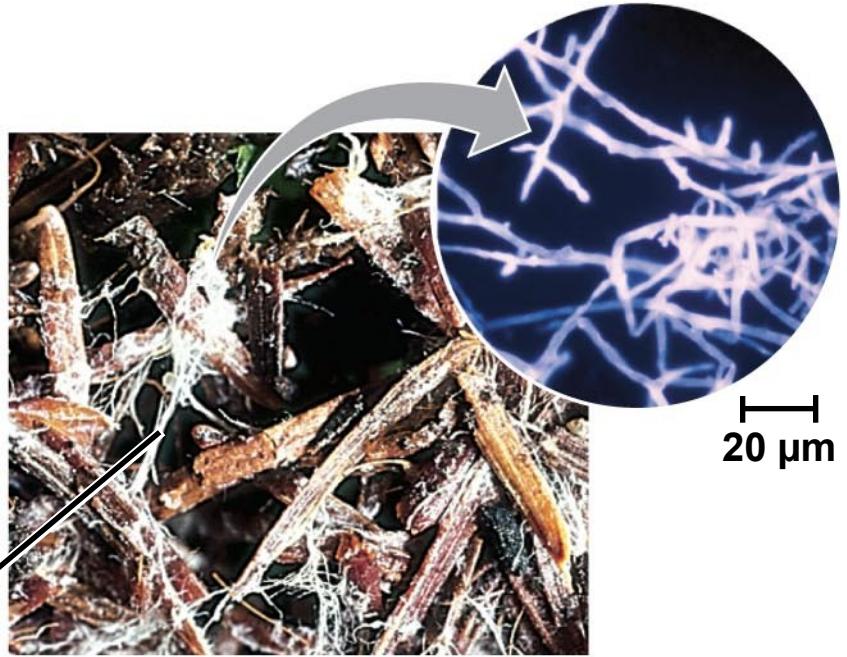
Multicellular Fungus: hyphae ...

Reproductive structure

Hyphae

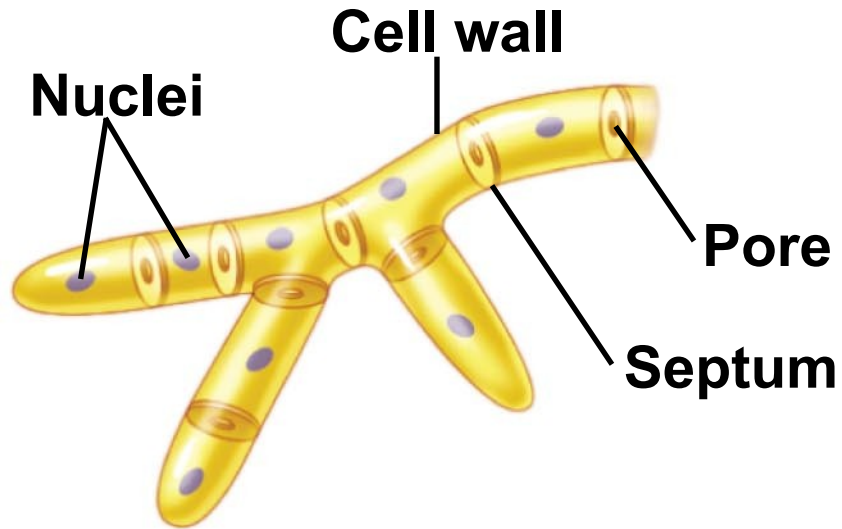
Spore-producing structures

Mycelium



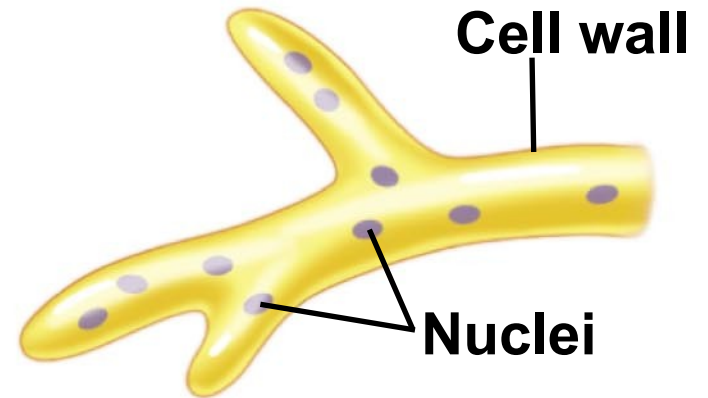
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- ***Septate fungi*** - Some fungi have **hyphae** divided into **cells** by **septa**, with pores allowing cell-to-cell movement of organelles.
 - ***Coenocytic fungi*** lack septa.
 - Some unique fungi have ***specialized hyphae*** called ***haustoria*** that allow them to ***penetrate the tissues of their host***.

Two forms of hyphae



(a) **Septate** hypha

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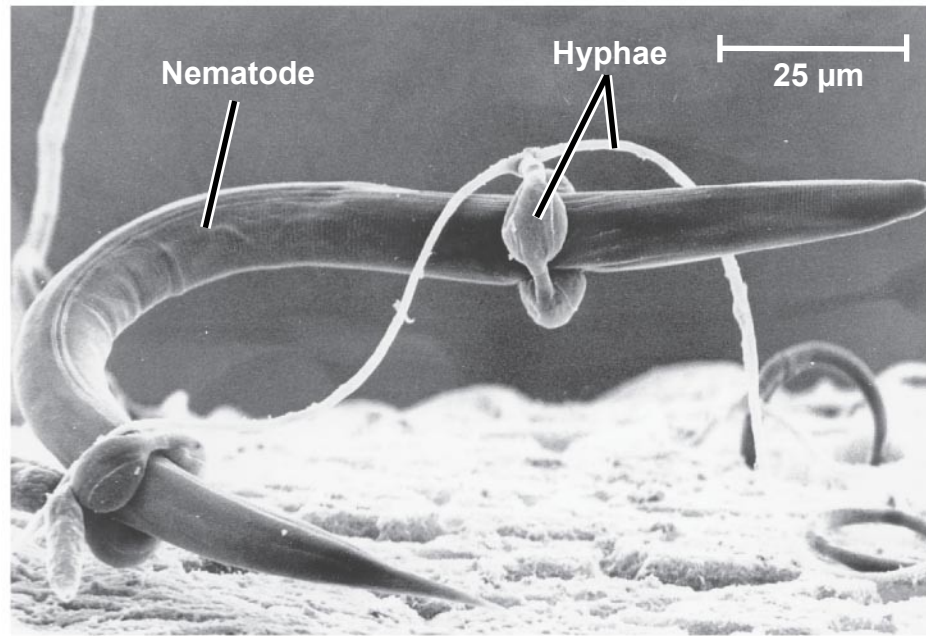


(b) **Coenocytic** hypha

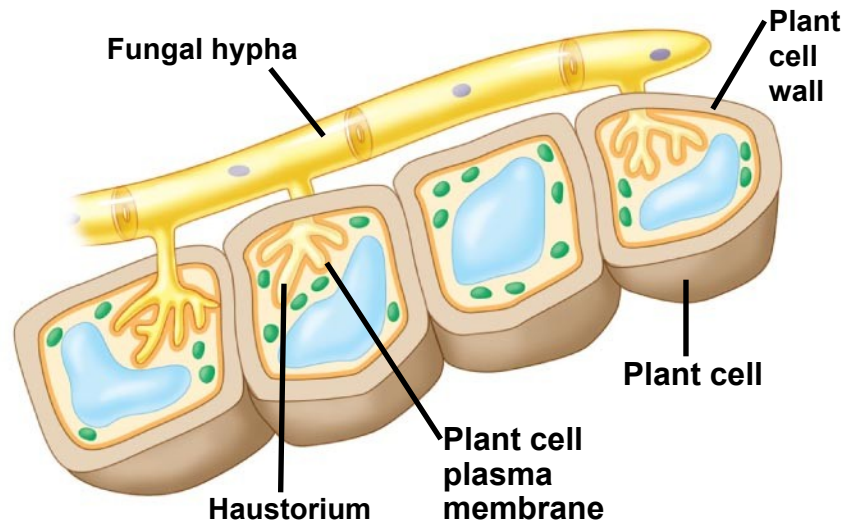
Specialized Hyphae in Mycorrhizal Fungi ++

- ***Mycorrhizae*** ++ are mutually beneficial symbiotic relationships between *fungi and plant roots*.
- **Ectomycorrhizal fungi** form sheaths of hyphae over a root and also grow into the extracellular spaces of the root cortex.
- **Arbuscular mycorrhizal fungi** extend hyphae through the cell walls of root cells and into tubes formed by invagination of the root cell membrane.

Fungus - have specialized hyphae



(a) **Hyphae adapted for trapping and killing prey**

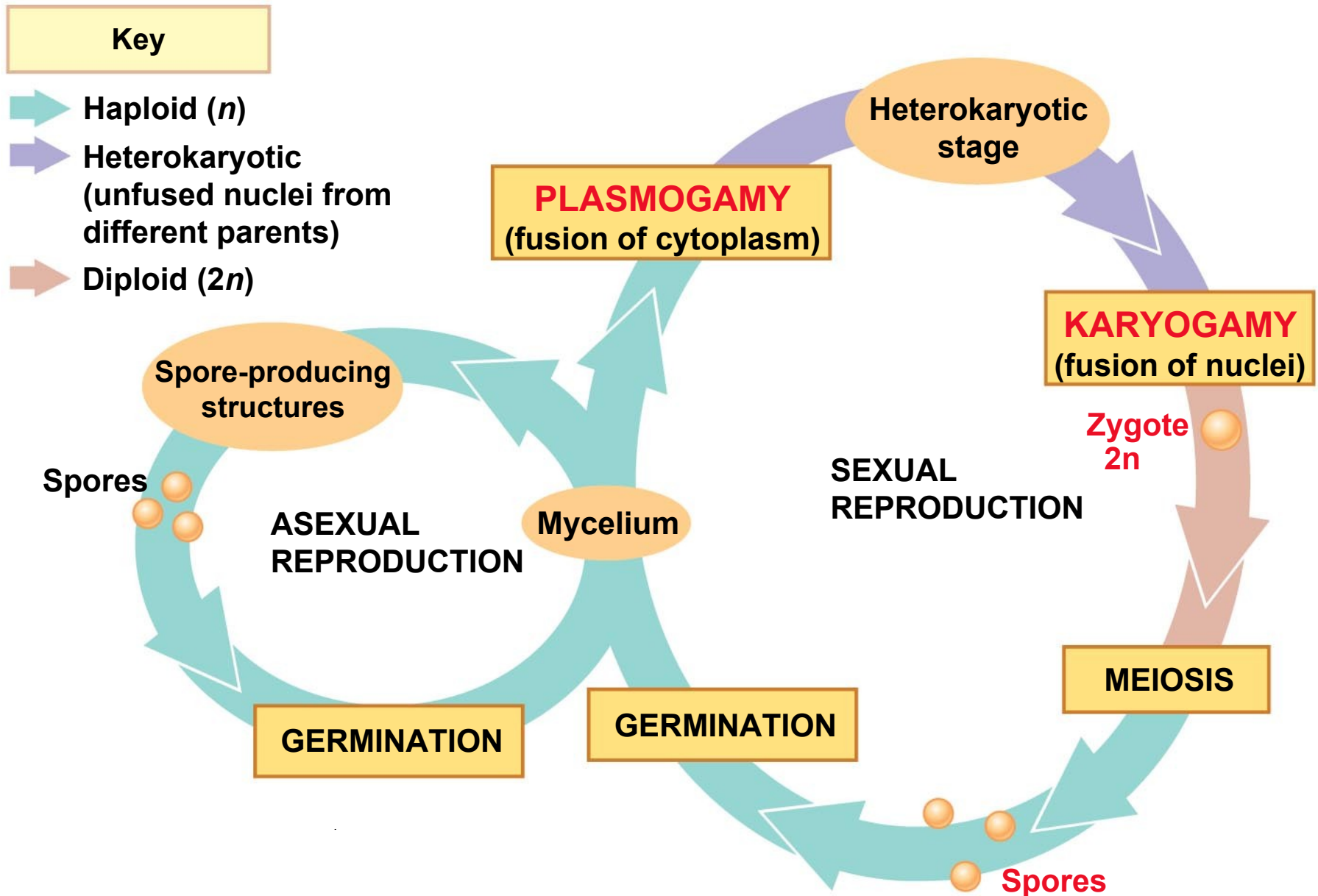


(b) **Haustoria - penetrate cell walls of plants ++ or +-
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Fungi produce spores through sexual or asexual life cycles

- Fungi propagate themselves by producing vast numbers of spores, either sexually or asexually.
- Fungal nuclei are normally haploid.
- Sexual reproduction requires the fusion of hyphae from different mating types.
- Fungi use *sexual signaling molecules* called *pheromones* to communicate their mating type.

Life Cycle of Fungi



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- ***Plasmogamy*** is the *union of two parent hyphae*, mycelia.
 - In most fungi, the *haploid nuclei* from each parent do not fuse right away; they *coexist* in the hyphae, this mycelium is called a **heterokaryon**.
 - In some fungi, the haploid nuclei pair off two to a cell; such a mycelium is ***dikaryotic***.
 - During ***karyogamy***, the haploid *nuclei fuse*, producing diploid cells: $n + n = 2n$.

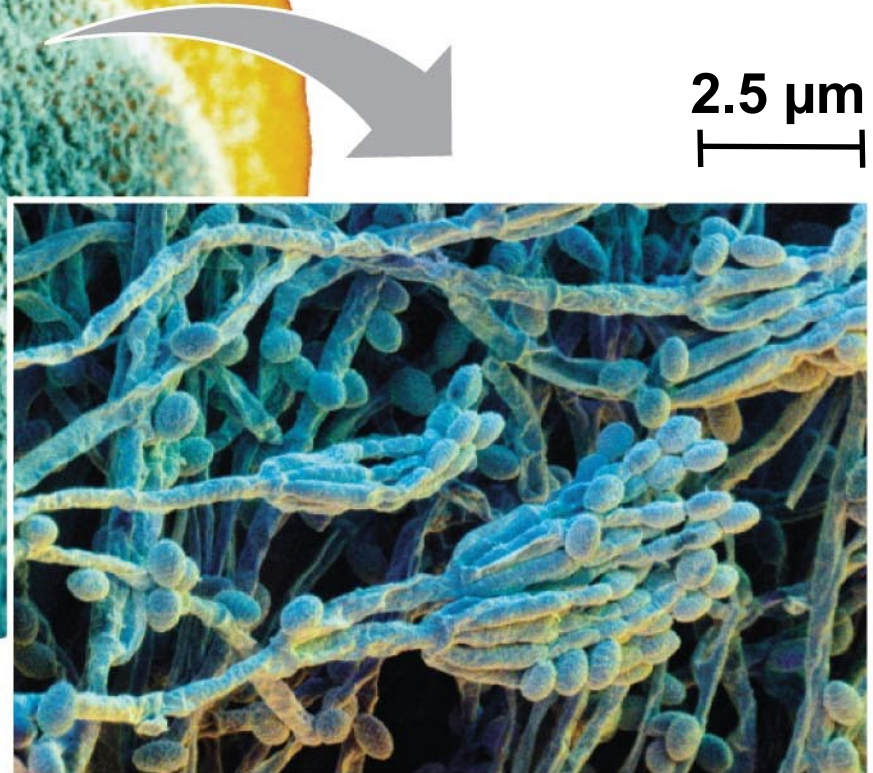
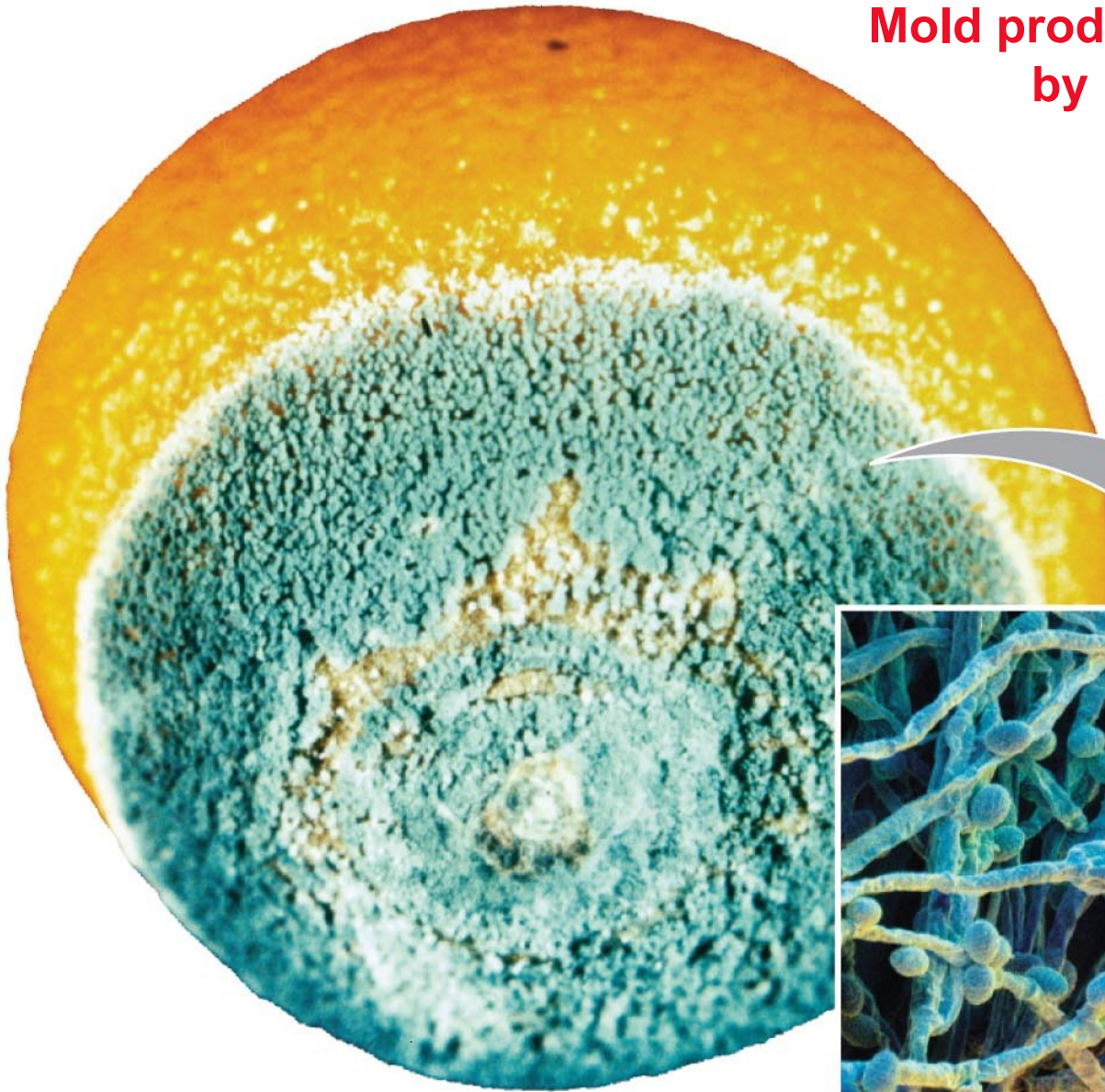
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- Hours, days, or even centuries may pass before the occurrence of *karyogamy*, *nuclear fusion*.
 - The diploid phase is short-lived and undergoes meiosis, producing haploid spores.

Asexual Reproduction

- In addition to sexual reproduction, many fungi can reproduce asexually.
- **Molds** produce *haploid spores* by *mitosis* and form *visible mycelia*.
- Other fungi that can reproduce asexually are yeasts, which inhabit moist environments.
- Instead of producing spores, *yeasts* reproduce *asexually by budding*: simple cell division and pinching of “bud cells” from a parent cell.

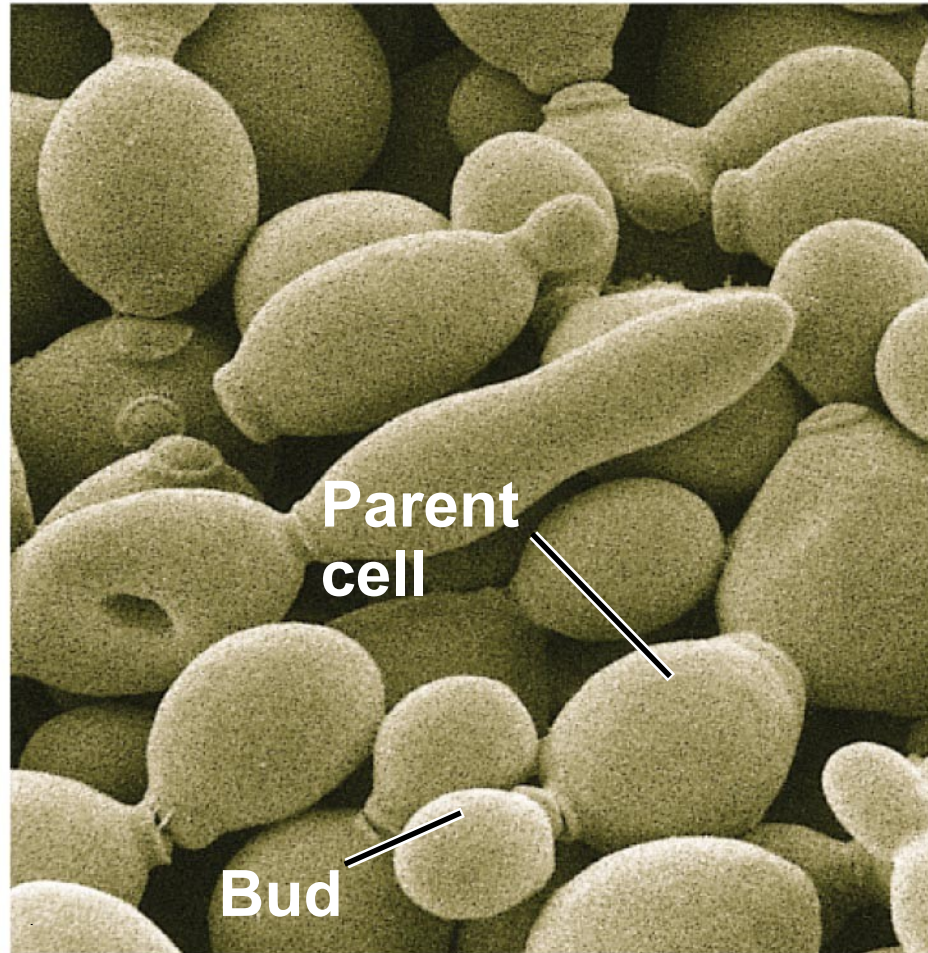
Penicillium: a mold that decomposes food.

**Mold produce spores asexually
by mitosis.**



The **yeast** *Saccharomyces cerevisiae* in several stages of **asexual** reproduction by **budding**.

10 μm

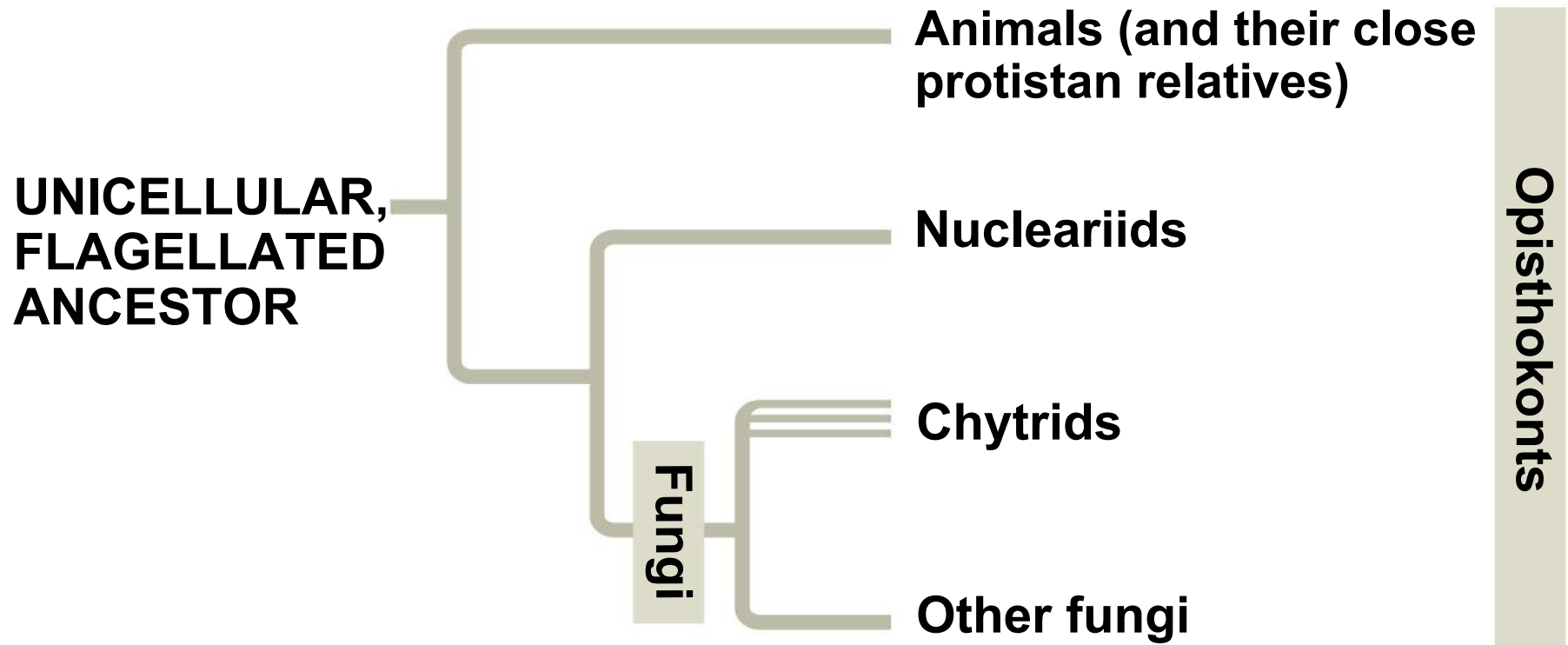


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Concept 31.3: The ancestor of fungi was an aquatic, single-celled, flagellated protist

- Fungi and animals are more closely related to each other than they are to plants or other eukaryotes.
- DNA evidence suggests that fungi are most closely related to unicellular **nucleariids** while animals are most closely related to unicellular choanoflagellates.
- This suggests that *fungi and animals evolved from a common flagellated unicellular ancestor* and multicellularity arose separately in the two groups.
- The oldest undisputed fossils of fungi are only about 460 million years old.

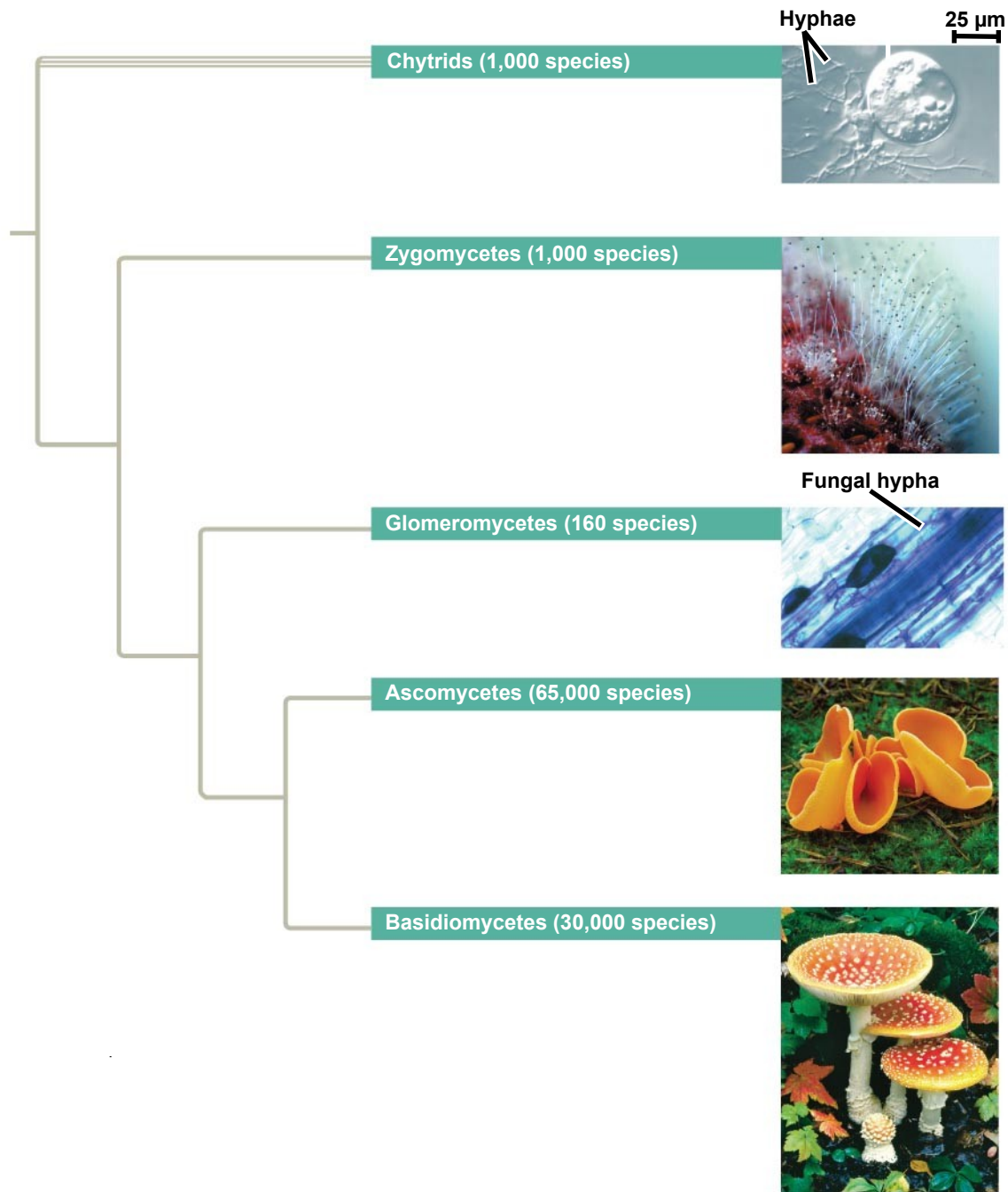
Fungi and their close relatives



The Move to Land

- *Fungi were among the earliest colonizers of land and probably formed mutualistic ++ relationships with early land plants.*
- **Chytrids** (phylum Chytridiomycota) are found in freshwater and terrestrial habitats.
- They can be decomposers, parasites, or mutualists.
- Chytrids are unique among fungi in having flagellated spores, called **zoospores**.

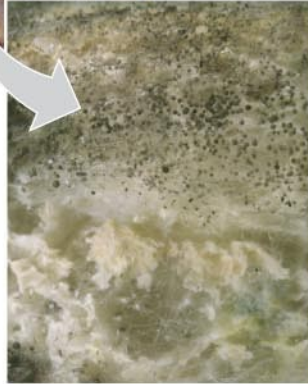
Fungus Diversity



Zygomycetes

- The **zygomycetes** (phylum Zygomycota) exhibit great diversity of life histories.
- They include fast-growing **molds**, parasites, and commensal symbionts.
- The zygomycetes are named for their sexually produced **zygosporangia**.
- Zygosporangia, which are resistant to freezing and drying, can survive unfavorable conditions.

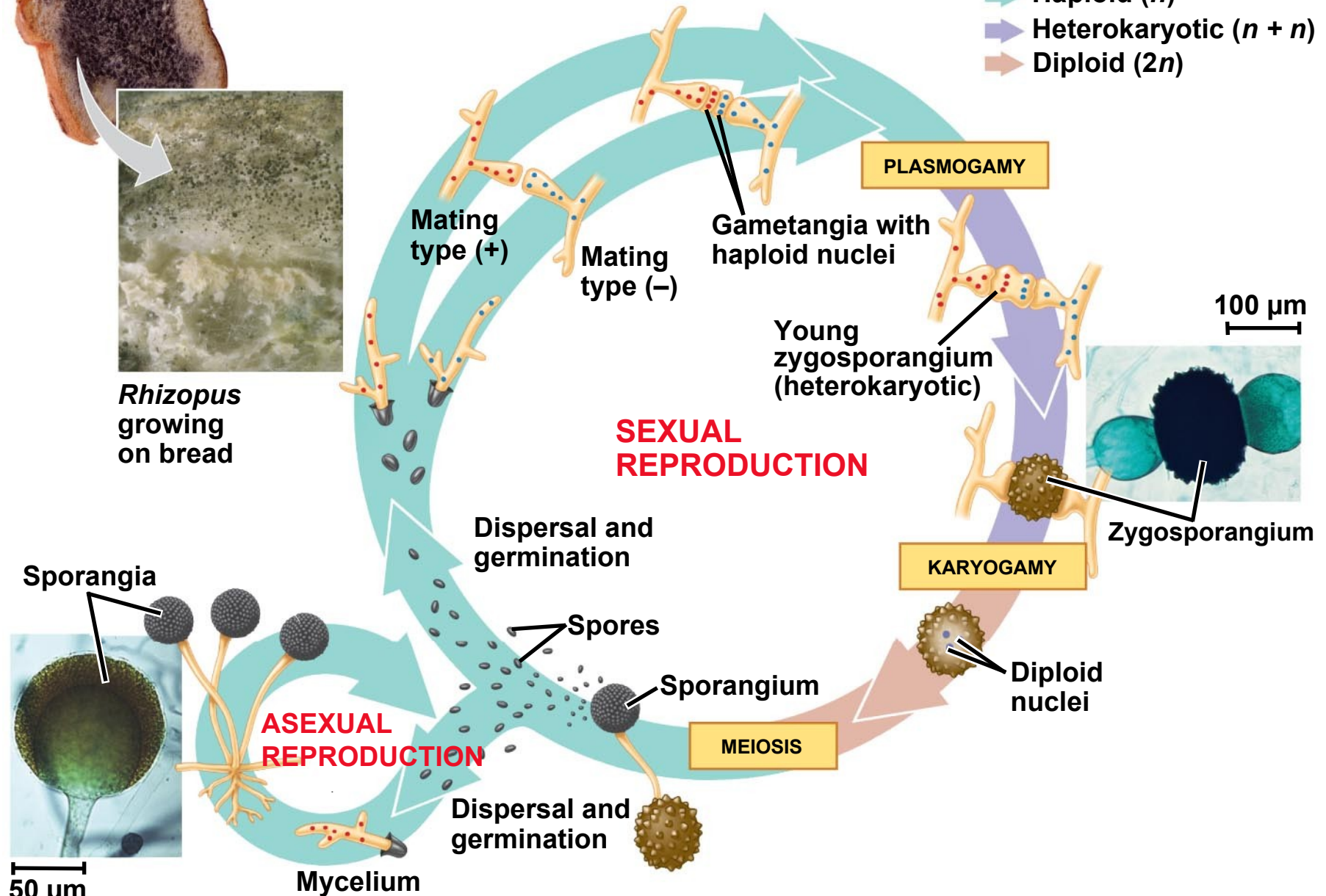
Life Cycle of the zygomycete *Rhizopus* black bread mold



Rhizopus
growing
on bread

Key

- ➔ Haploid (n)
- ➔ Heterokaryotic ($n + n$)
- ➔ Diploid ($2n$)



Some zygomycetes, such as *Pilobolus*, can actually “aim” their sporangia toward conditions associated with good food sources.



0.5 mm

Ascomycetes

- **Ascomycetes** (phylum Ascomycota) live in marine, freshwater, and terrestrial habitats.
- The phylum is defined by production of sexual spores in saclike **asci**, usually contained in fruiting bodies called **ascocarps**.
- Ascomycetes are commonly called **sac fungi**.
- Ascomycetes vary in size and complexity from **unicellular yeasts** to elaborate cup fungi and morels.

Ascomycetes - sac fungi

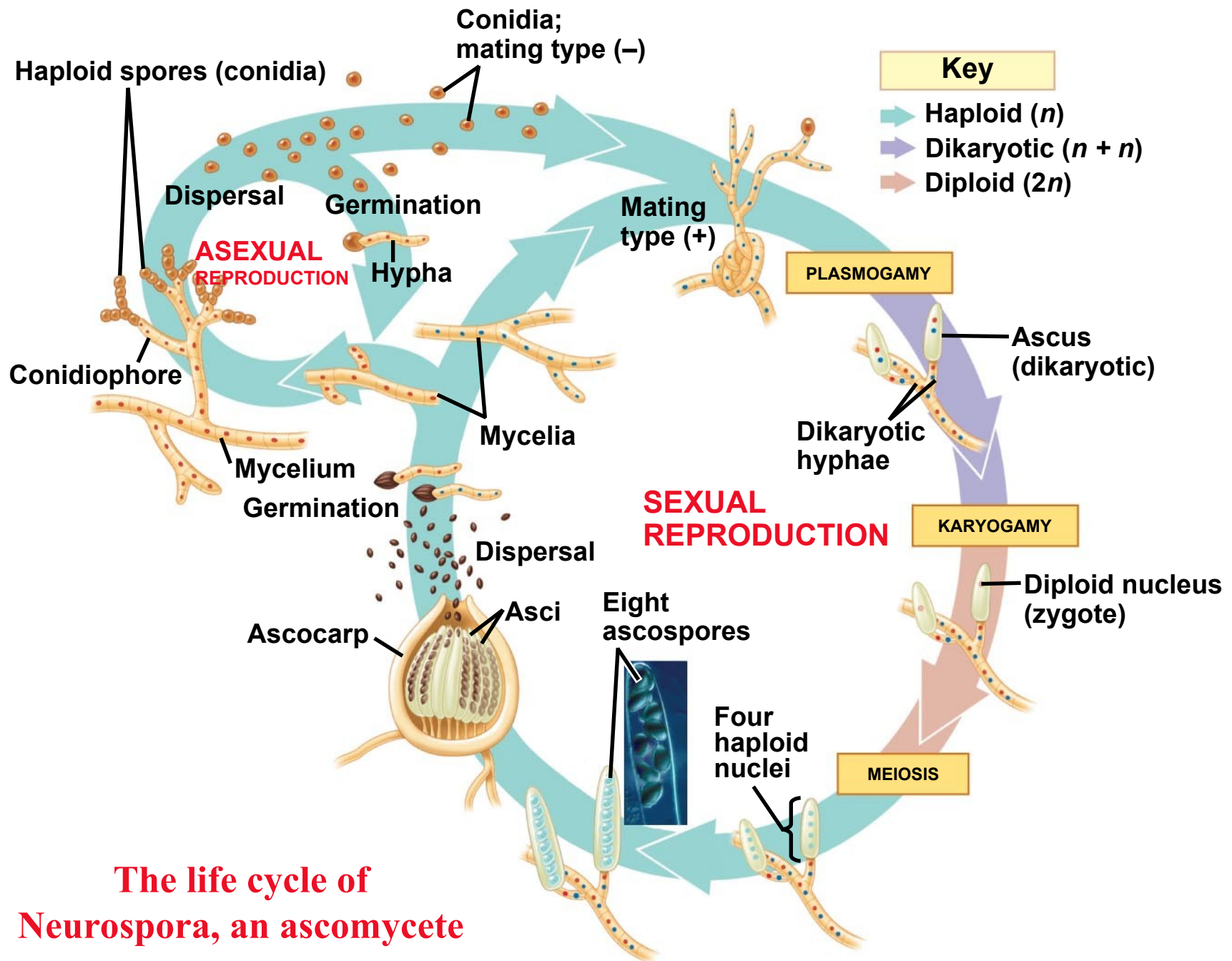
Morchella esculenta,
the tasty morel



Tuber melanosporum, a truffle



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- Ascomycetes include plant pathogens, decomposers, and symbionts
 - Ascomycetes reproduce asexually by enormous numbers of *asexual spores* called *conidia*.
 - Conidia are not formed inside sporangia; they are produced asexually at the tips of specialized hyphae called conidiophores.
 - *Neurospora* is a model organism with a well-studied genome.



Basidiomycetes

- **Basidiomycetes** (phylum Basidiomycota) include *mushrooms*, puffballs, and shelf fungi, mutualists, and plant parasites.
- The phylum is defined by a clublike structure called a **basidium**, a transient diploid stage in the life cycle.
- The *basidiomycetes* are also called *club fungi*.

- ▶ Maiden veil fungus (*Dictyophora*), a fungus with an odor like rotting meat



Basidiomycetes club fungi



- ◀ Puffballs emitting spores

- ▼ Shelf fungi, important decomposers of wood



Concept 31.5: Fungi play key roles in nutrient cycling, ecological interactions, and human welfare

- Fungi interact with other organisms in many ways.
- Fungi are efficient *decomposers*. They perform essential *recycling of chemical elements between the living and nonliving world*.
- Fungi form *mutualistic relationships* with plants, algae, cyanobacteria, and animals.
- All of these relationships have profound ecological effects.

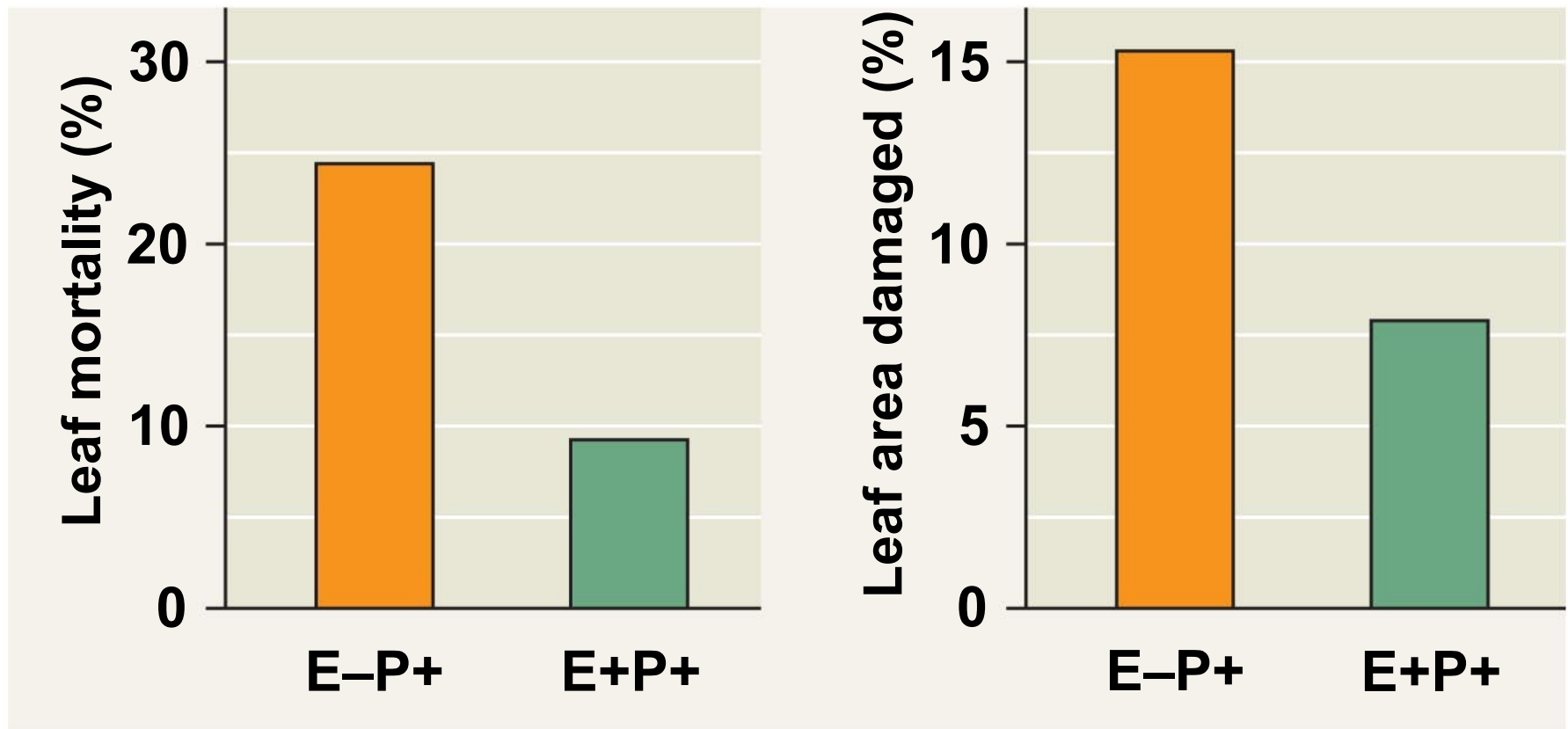
Fungus-Plant Mutualisms ++

- ***Mycorrhizae*** live ++ symbiotically in ***plant roots*** and are enormously important in natural ecosystems and agriculture.
- Plants harbor harmless **symbiotic endophytes** that live **inside leaves** or other plant parts ++
- Endophytes **make toxins** that deter herbivores and **defend against pathogens**.

Do endophytes benefit a woody plant?

RESULTS

- Endophyte not present; pathogen present (E–P+)
- Both endophyte and pathogen present (E+P+)



Fungus-Animal Symbioses

- Some fungi *share* their *digestive services* with animals.
- These fungi help break down plant material in the *guts of* cows and other *grazing mammals*.
- Many species of *ants* and *termites* use the digestive power of fungi by raising them in *“farms.”*

Fungus-gardening insects: Leaf-cutting ants depend on fungus to digest plant material to a form the insects can use for their nutrition ++.



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Lichens

- A *lichen* is a ++ symbiotic association between a *photosynthetic microorganism* and a *fungus* in which millions of photosynthetic cells are held in a mass of fungal hyphae.
- The *fungal* component of a *lichen* is most often an ascomycete.
- *Algae or cyanobacteria* occupy an inner layer below the lichen surface.

Variety of lichens + +

▼ A fruticose (shrublike) lichen

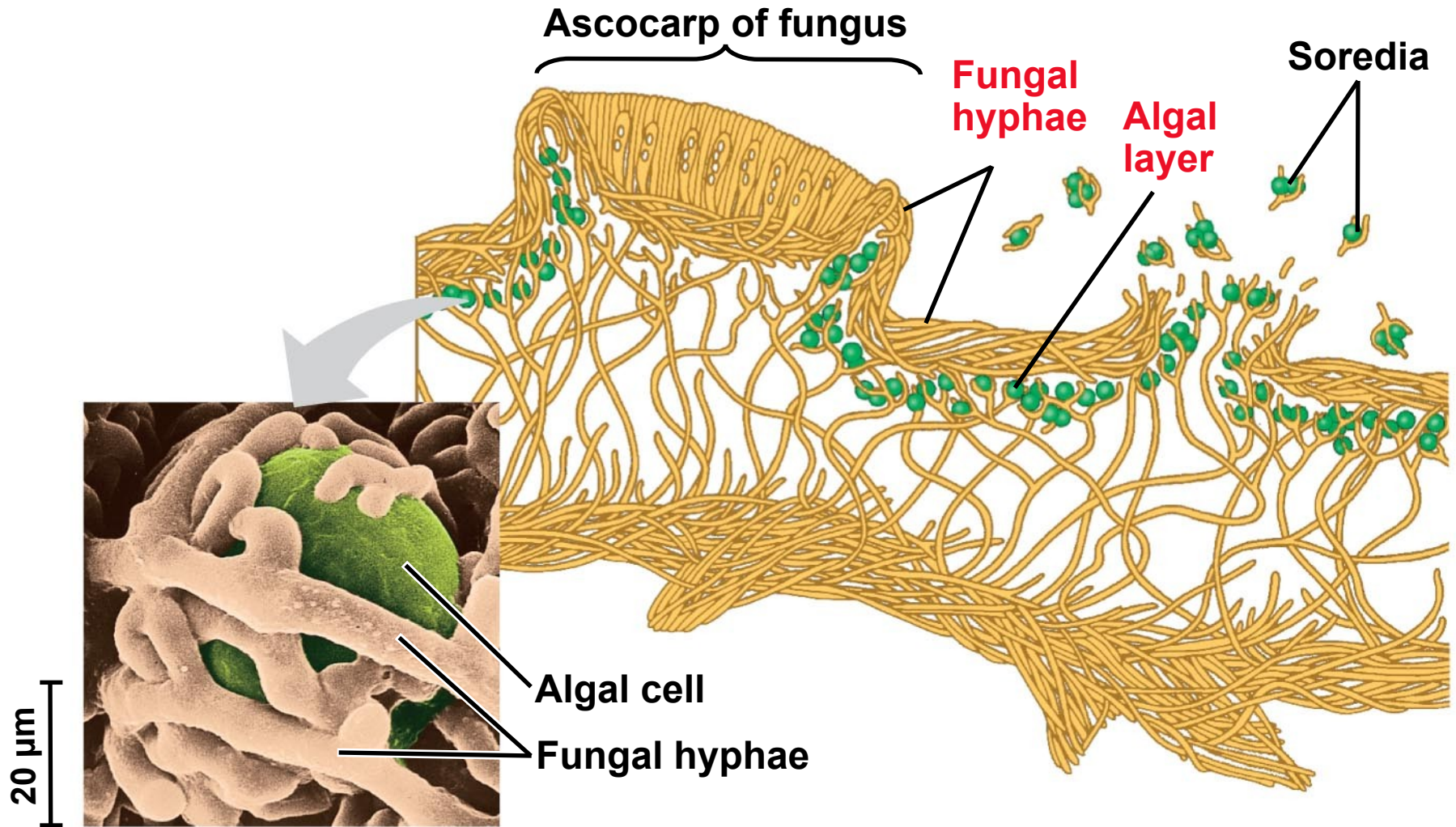


◀ Crustose
(encrusting)
lichens

▼ A foliose
(leaflike)
lichen



Anatomy of a common fungal ascomycete **lichen**



Lichen ++

- The *algae provide carbon compounds*, cyanobacteria provide organic nitrogen, and *fungi provide the environment for growth*.
- The fungi of lichens can reproduce sexually and asexually. Asexual reproduction is by fragmentation or the formation of **soredia**, small clusters of hyphae with embedded algae.
- *Lichens are important pioneers on new rock and soil surfaces -- pioneer organisms in ecological succession.*
- Lichens are sensitive to pollution, and their death can be a warning that air quality is deteriorating.

Fungi as Pathogens + -

- About 30% of known fungal species are *parasites or pathogens*, mostly on or in plants.
- Some fungi that attack food crops are toxic to humans.
- Animals are much less susceptible to parasitic fungi than are plants.
- The general term for *a fungal infection* in animals is *mycosis*.

Fungal Diseases in Plants



(a) Corn smut on corn



(b) Tar spot fungus on maple leaves



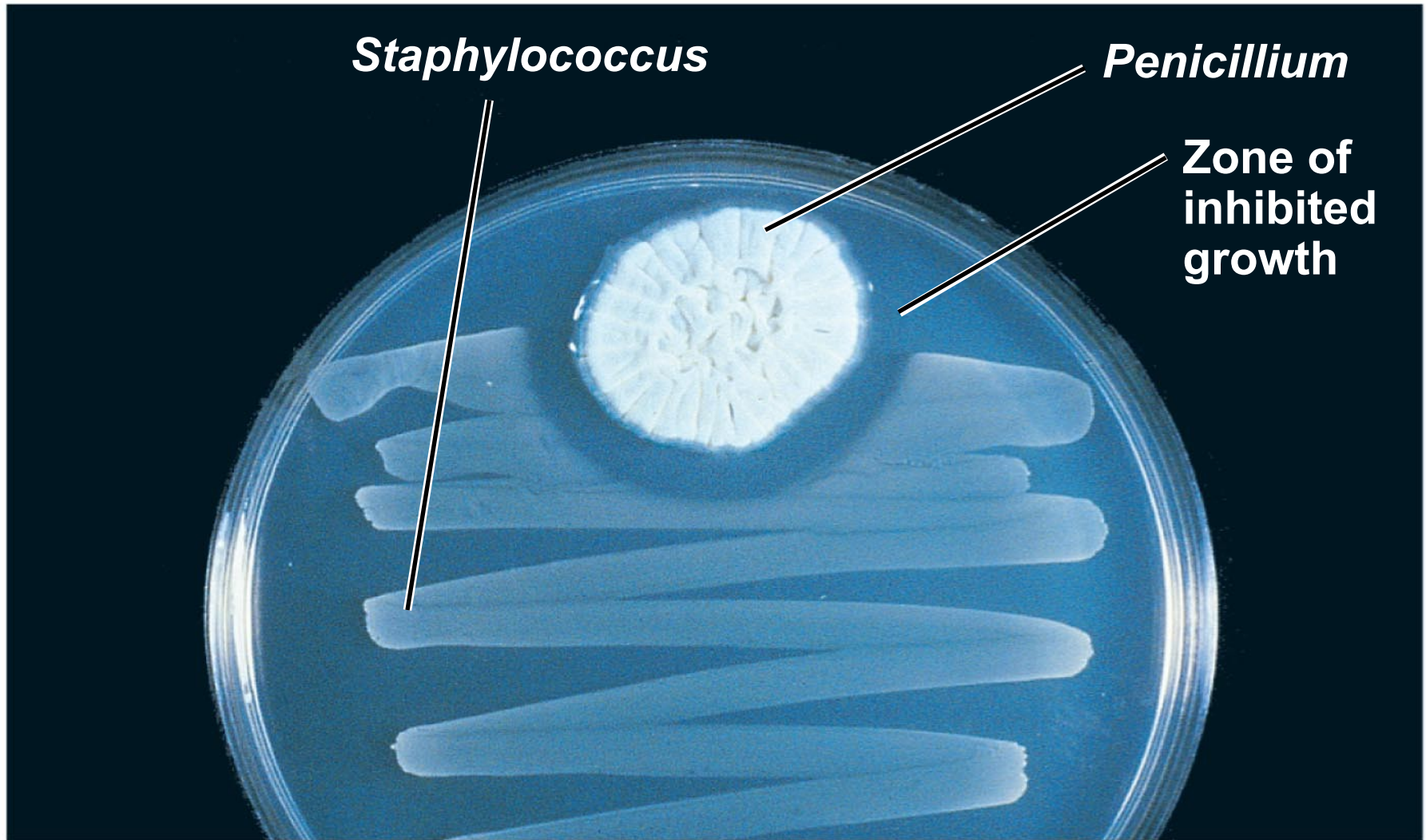
(c) Ergots on rye

Practical Uses of Fungi

- *Food*: Humans eat many fungi and use others to make cheeses, alcoholic beverages, and bread.
- Some fungi are used to produce *antibiotics* for the treatment of bacterial infections, for example the ascomycete *Penicillium*.
- Genetic research on fungi is leading to applications in *biotechnology*:
 - For example, insulin-like growth factor can be produced in the fungus *Saccharomyces cerevisiae*.

Penicillium: Fungal production of an Antibiotic

The mold penicillium produces an antibiotic that inhibits bacteria growth resulting in a clear area between the mold and the bacteria



Review

Fungal Phylum	Distinguishing Features of Morphology and Life Cycles	
Chytridiomycota (chytrids)	Flagellated spores	
Zygomycota (zygote fungi)	Resistant zygosporangium as sexual stage	
Glomeromycota (arbuscular mycorrhizal fungi)	Form arbuscular mycorrhizae with plants	
Ascomycota (sac fungi)	Sexual spores (ascospores) borne internally in sacs called asci; ascomycetes also produce vast numbers of asexual spores (conidia)	
Basidiomycota (club fungi)	Elaborate fruiting body (basidiocarp) containing many basidia that produce sexual spores (basidiospores)	

You should now be able to:

1. List the characteristics that distinguish fungi from other multicellular kingdoms.
2. Discuss mycorrhizal fungi.
3. Describe the processes of plasmogamy and karyogamy.
4. Describe the evidence that multicellularity evolved independently in fungi and animals.

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5. Describe the life cycles of *Rhizopus stolonifer* and *Neurospora crassa*.
 6. Distinguish among zygomycetes, ascomycetes, and basidiomycetes.
 7. Describe some of the roles of fungi in ecosystems, lichens, animal-fungi mutualistic symbioses, food production, and medicine and as pathogens.